DYLAN CONARD

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Princeton University, **A.B. in Computer Science, GPA: 3.72** *Expected Graduation: May 2027*
Minor: Environmental Studies

Research Experience

Meteorology Undergraduate Researcher at University of Reading April 2026 – Aug. 2026

- Assessed the performance of a novel ML feature selection technique (CRO-Spatio) for predicting annual tropical cyclone genesis predictions on a global and basin-wide scale through regression metrics and feature attribution analysis.
- Extended the code's functionality to allow for more diverse training datasets by modularizing the temporal and spatial input features.
- Investigated the feasibility of including a more physically informed algorithm to allow for a more dynamical model.

Civil and Environmental Engineering Undergraduate Researcher at Princeton University Feb. 2024 – Aug. 2025

- Assessed the performance of new human vulnerability metrics for extreme weather hazards in different regional climates across the US, identifying discrepancies of over 7x more extreme heat events in the Northeast for Wet-Bulb Globe Temperature.
- Cleaned and validated a 20-year, daily, county-level data set using NOAA weather archives to ensure the conclusions were reliable.
- Scaled the project's analytical scale from county to regional patterns, enabling for more robust geographical comparisons.

Spatial XAI-GPI – A Spatially-Informed Explainable Genesis Potential Index (Senior Thesis Project) Aug. 2026 – Present

- Integrating dimensionality reduction techniques and an evolutionary optimization method for feature selection into a deep neural network pipeline, this app aims to be used with observed and GCM data to study tropical cyclone genesis and seed development.
- Tests perturbations of GCM simulations to analyze future conditions and understand the variables relative feature importance.

WindVision – Using Augmented Reality Visualize Air Flow (Junior Research Project) Jan. 2026 – May. 2026

- Utilized AI vision powered room scans on augmented reality headset to simulate and visualize airflow within a real-world room to help better understand and communicate the importance of window-based ventilation techniques for cooling.
- Integrated customizable, interactable objects to help plan out methods to increase air flow throughout the room within the app.

Work Experience

SIFP Head Fellow – *Princeton Emma Bloomberg Center for Access and Opportunity* Jan. 2026 – Present

- Advises 31 Princeton students by organizing weekly professional and academic mentorship meetings while also setting up monthly 1-on-1 meetings to understand how to address the needs of each mentee.
- Receives intensive and inclusive leadership training experience, including for instance facilitation, communication, and debriefing skills to better support the mentees.

ACT Tutor – *Princeton Tutoring LLC* July 2024 – Present

- Identified 11 high school students' opportunities for growth and created personalized short- and long-term plans to achieve their ACT goals through regimented online or in-person tutoring sessions that result in +10 percentile score increase.
- Instructed students in evidence-based ACT strategies that emphasize problem-solving, critical reasoning, and structured study habits, while also strengthening organization and time management habits.

Relevant Coursework

Tropical Cyclones and Associated Impacts Jan. 2026 – May. 2026

- Storm surge modeling, inland flooding effects, wind damage metrics, energy infrastructure, public health, and risk management.

Climate and Weather: Order in the Chaos Sep. 2025 – Dec. 2025

- Energy and water cycling, oceanic and atmospheric circulation and fluxes, solar heating, ENSO, and statistical analysis.

Intro to Machine Learning Sep. 2025 – Dec. 2025

- Classification and regression, SVMS, clustering, dimensionality reduction, deep neural networks, and reinforcement learning.

Independent Projects

Machine Learning Prediction Algorithm for Audio Features Aug. 2026 – Present

- Applies correlation analysis and regression models to capture linguistic signals within music criticism reviews that correspond to computationally measured audio properties to examine whether music criticisms can be acoustically measured.

TigerConnect – A Social Scheduling App for Princeton Students Jan. 2026 – May. 2026

- Developed a social scheduling app aimed to boost connection and reduce the friction between scheduling meeting times by integrating a personal calendar, automatic status indication, messaging system, and personalized suggested meeting times.

Machine Learning Fraud Detector Nov. 2024 – Dec. 2024

- Implemented a lightweight ML model using decision stumps and multiplicative weights to detect fraudulent credit card transactions within large sets of data.

Skills and Extracurriculars

Programming: Python, Java, C, GoLang, Bash, Qiskit, PyTorch, Seaborn/Matplotlib, SHAP, NumPy, Pandas, Scikit, XArray

Affiliations: ASL Club Officer, TiGHAs IM Soccer Co-Captain, Club Rock Climbing, Whig-Clio, Cap & Gown Club